Player Skill Set Grouping and Price Insights for Indian Premier League (IPL) Auction



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Abstract

This project deals with a popular sport played in England, South Africa, Australia, New Zealand, and Asian countries such as India, Pakistan, Bangladesh and Afghanistan; cricket. There are 3 formats of cricket, of which the shortest played format is called Twenty20. The Indian Premier League is a professional cricket tournament that consists of 8-9 franchises owned by famous business magnates and celebrities, that recruit professional players and new budding talents to play for their franchises.

The general game of cricket is played between 2 teams. Each team consists 11 players who are shortlisted among the squad of 20-25 players that is picked by each franchise in an auction. There are two aspects to the game; batting and bowling. The batting team scores as many runs as they can in 20 overs. Each over consists of six balls each bowled by a bowler; so 120 balls in total. The bowling team tries to get the batsmen out by either catching the ball when they hit it, or hitting the stumps when the batsman misses the ball or is out of his crease running. After 20 overs, both the teams switch roles. The original bowling team tries to bat the 20 overs to beat the other team's score. At the end of the game, the team which scores more runs wins the match.

Cricket is very similar to baseball, therefore is an intense game that demands the most skilled players. IPL is a franchise cricket league that is tied to millions of dollars of investment in which each of the franchises play 2 games against each other in the league stage of the tournament. The top 4 teams of the league stage battle each other in the playoffs, and 2 teams play the final. The winner of the final, is declared to be the winner of the tournament and is entitled to a lumpsum of prize money.

To win the tournament, franchises must ensure that they pick the best squad possible in all departments while they compete in the auction, where they will bid for players. This project aims to ease the decision making process through insights where we will group players with similar skillsets and records, in addition to predicting prices for budgeting purposes. Each team has a capped total amount of how much they can spend in total.

We will be looking at player statistics from various sources (datasets that contain information about their performance in the previous editions of the Indian Premier League) and will be exploring individual players and differences/similarities between them. Also, we have applied machine learning based algorithms that predicts the cost at which a player can be sold in the Indian Premier League Auction. In this we will use Data Analysis Methods: Principal Component Analysis, K-means Clustering, Charts & Plots. For Auction Price Prediction Method we will make use of Regression Trees, K-nearest Neighbors(KNN), Linear Regression, Support Vector Regression, Random Forest Regressor.

Problem Statement

The Indian Premier League (IPL) is a professional Twenty20 cricket league that was started in 2008 in India.[1] The IPL is the most-attended cricket league in the world and in 2019 was ranked third by average attendance among all sports leagues.[2] In 2019 the brand value of IPL was $6.3 Billion. [3] The participating IPL franchises form their teams by conducting an auction before the tournament from a pool of national and international cricket players of different countries. As per IPL regulations, each franchise can spend a maximum of US $12.5m on their team salaries [4]. To select the correct player and predict the auction price of a batsman there are several important parameters to consider like "Batting average", "Batting strike rate", "Consistency in Scoring Runs" etc. Similarly, to predict the price of a bowler there are several important parameters to consider like "Economy rate", "Number of Wickets", etc. At present, the decision of choosing to bid for a player is done by eight representatives from each franchise. Each has different expertise and contributes to the final decision making. This makes it difficult to make decisions as there might be bias, and too many ideas on the table. Furthermore, pressure situations and hurried decisions which need to be made at the auction can lead to costly mistakes and can ultimately lead to slim chances at member winning the high stakes tournament. Our proposed predictions made by machine learning models and player skillset analysis insights makes the decision making more accurate, and well prepared for all possible scenarios that may arise. We can group players (batsmen and bowlers) based on similar skill sets and group similar players according to their past records. We are also able to predict the price of a player based on his past performance. Our model helps the bidder realize the exact range of the price the player is expected to be sold at, thereby enabling them to plan and adjust their budget. At present the accuracy with which auctioneers are able to evaluate a player is dependent on the experience in evaluating that the 8 members of the bidding team have. We aim to take the help of statistical analysis to eliminate some of the bias that the franchise members might bring to a table, and give new insights that might not have been considered before.

Data Source

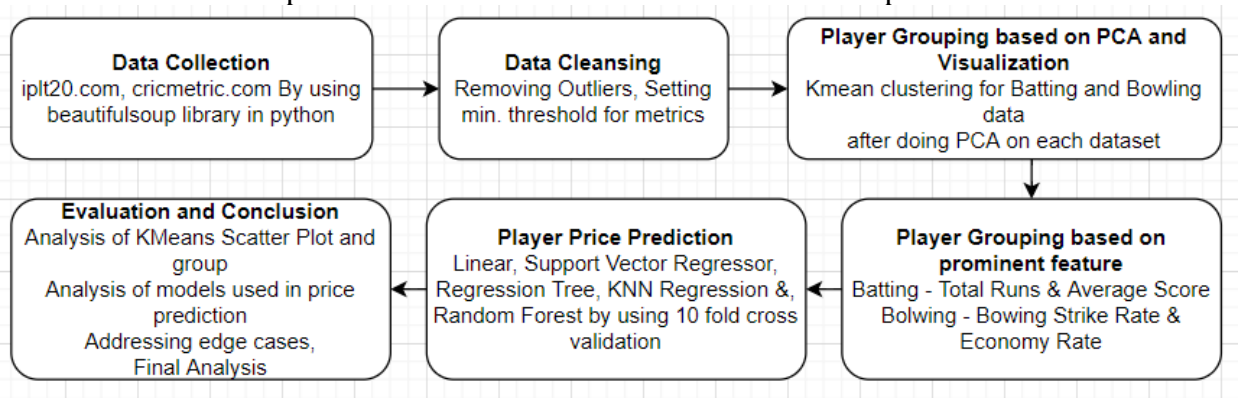
We are using player data of 2018–2021 IPL seasons from the following website iplt20.com and IPL Auction price data will be used from website cricmetric.com.

Methodology

Players would be grouped based on different metrics such as strike rate, economy rate, number of wickets, total runs scored, etc. The parent datasets would require manipulation to extract this information and group players so that franchises will be aware of players with skill sets. The above metrics will be used throughout our analysis of players. The steps to arrive at our grouping and price prediction insights are mentioned below:

Solution Steps -

We have divided the problem statement into below workflow steps:



Flow Diagram - Solution

Data Collection -

The data would be scraped using the beautiful soup library from iplt20.com where we can find stats for batting and bowling. The features that will be considered for the batsmen are Strike Rate, Number of Runs Scored , Matches Played, Number of Sixes, Number of Fours, Number of 50s and Number of 100s, Batting Average. The features that will be considered for bowling are Wickets, Economy Rates, Bowling Strike Rate, Innings, Overs, Number of Four and Five Wicket Hauls, and Number of Dot Balls Faced. The Salaries were scraped from cricmetric.com.

Batting Data		Bowling Data	
Column	Description	Column	Description
Player	Name of player	Player	Name of player
Mat	Number of match played	Mat	Number of match played
I or Inns	Number of inning players played	I or Inns	Number of inning players played
R or Runs	Total Number of Run scored	Ov or O	Number of over bowled
B or BF	Total number of ball faced	Runs	Total number of runs conceded
Outs	Number of out	Wkts or W	Total wicket taken
Avg	Average Score	Avg	Average run conceding per wicket
SR	Strike Rate	Econ	Run conceded per over
HS	Highest Score	SR	Avg number of balls bowled per wicket taken
4s	Number of four hit	4W	Number of time 4 wickets taken
6s	Number of six hit	5W	Number of time 5 wickets taken
50	Number of times scored 50 run	4s	No of 4 run conceded (boundry)
100	Number of times scored 100 run	6s	No of 6 run conceded (boundry)
Salary	Salary of player	Dots	Total number balls bowled without run
Year	IPL Year	Salary	Salary of player
		Year	IPL Year

Data Cleaning and Outlier Removal -

Some basic cleaning has been incorporated to ensure we have reliable data. The datasets that were scraped were already ready for analysis. Some changes that were done was to remove commas from numeric types and convert to floating points and integers. We have also applied filters on the batting and bowling datasets before analysis to remove outliers and data that might skew our results.

For batting, the following filters were applied to remove outliers :

- Total Runs greater than 200,
- Average Score greater than 20
- Strike Rate greater than 101

For bowling, the following filters were applied to remove outliers:

- Minimum Number of Overs Bowled Must be greater than 8 (Ov>8)
- The player must have featured in atleast 3 games. (Innings>3)
- The player must have taken greater than 8 wickets

These above-mentioned conditions help us filter out those batsmen who bowl part time and those bowlers who have not played much of an influential role (Their statistics may be skewed, which will effect the clustering result.)

Player Grouping -

We performed K-means on each individual feature such as strike rate, batting average, number of runs scored for batsman. For bowlers, we performed K-means on economy rates, number of wickets, bowling strike rate, number of four wicket hauls, and number of 5 wicket hauls. This gives us a detailed insight into grouping players into clusters based on each feature to identify players with similar skill sets. For overall grouping of players we performed Principal Component Analysis on all the features and take the top 2 components. Kmeans has been performed on the resultant components to group similar players on overall stats for both batting and bowling. Important features were recognized for both batting and bowling and K-means was performed on the top 2 features for each as well.

Player Grouping Batting -

1. Standardize (Scaled) the range of all input data variables
2. Applied Dimensionality Reduction techniques such as Principal Component Analysis to preserve the variation of data and scale down to the top 2 components.
3. Used elbow method to determine the Optimal number of clusters in a data set
4. Created 6 groups of Batsman who have similar skill set using KMeans algorithm
5. Created another 6 group of Batsman by just using two input variable “Total Number of Runs” and “Average Score”

Player Grouping for Bowling-

1. Dropped irrelevant features such as Overs Bowled, Matches Played, and Innings as the players are required to be categorized based on skill and not experience.
2. Standardized (Scaled) all the features to have equal representation during K-means clustering analysis.
3. Applied Dimensionality Reduction techniques such as Principal Component Analysis to preserve the variation of data and scale down to the top 2 components.
4. Leveraged the Elbow method to identify the optimal number of clusters to use.
5. Applied K-means clustering on the data to group players according to all the attributes.
6. After applying clustering, we applied some basic feature selection methods after generating the correlation matrix. Two of the most influential features are Bowling Strike Rate (SR) and Economy Rate (Economy). I performed K-means clustering on the two attributes.

Auction Price Prediction -

Now that we have clusters of players available with similar skill sets, we would also provide insights into the likely auction price of each player, for budgeting purposes. We have considered the 2020,2019 and 2018 editions of the IPL and data from those editions for training and validation. Cross-validation techniques were leveraged for training and validation.

We created correlation matrices for batting and bowling, and identified the most influential features. These features were used for training and validating our models.

The unnecessary features were removed and different models such as classic regression, regression trees, SVM regression, KNN Regression, and Random Forest Regression would be run on the data.

The accuracy of the model will be evaluated on the basis of mean squared errors produced from the predicted salaries. The price prediction methodology followed for both batting and bowling is mentioned below:

Batsmen Price Prediction -

1. After getting data and cleaning mentioned in above steps, we identified the batsman by adding filter (*Total Runs* greater than 200, *Average Score* greater than 20 and *Strike Rate* greater than 101)
2. Standardize(Scaled) the range of all input data variables
3. Performed Linear Regression using 10 fold cross validation
4. SVR Regression model(non linear Kernel) using 10 fold cross validation
5. Regression Tree model using max depth =3
6. KNN Regression model after identifying optimal value of K
7. Predicted auction price of all players and calculated Means Square Error of all models

Bowling Price Prediction-

1. After basic data cleaning for the bowling data, we filtered the bowlers we want to consider in our analysis based on the following conditions:
    - Minimum Number of Overs Bowled Must be greater than 8 (Ov>8)
    - The player must have featured in atleast 3 games. (Innings>3)
    - The player must have taken greater than 8 wickets

    These above mentioned conditions help us filter out those batsmen who bowl part time and those bowlers who have not played much of an influential role (Their statistics may be skewed, which will effect the clustering result.)

2. Selected the most important features and removed the unnecessary features by generating the correlation matrix. We observed which features Salary is most correlated with. Features such as Overs, Wickets, Economy Rates, Bowling Strike Rate were used for training, validation and testing.
3. Scaled the features and dependent variable that we are predicting (Salary).
4. The bowling dataset is small as it has only about 129 players that are being considered. Therefore, we have leveraged using cross validation for training our data.
5. Performed Linear Regression using 10-fold cross validation.
6. Performed Support Vector Regression using 10-fold cross validation and non-linear rbf kernel.
7. Generated a regression tree with a max depth of 4 and predicted salaries.
8. Performed KNN regression. The optimal K was selected by running the KNN model and choosing the K value that gave the least error. The optimal K that was chosen was 20.
9. Predicted auction prices for all bowlers and calculated the Mean Squared Error for all the models.
10. Identified the Best Performing Model by comparing the respective MSE's for each model.

Evaluation and Final Results

As mentioned previously there are 2 aspects to providing insights for the IPL Auction; grouping players based on similar skill sets and predicting auction prices. It is important that the models we train and grouping must be accurate as heavy deviations can be catastrophic and can lead to a huge financial waste.

Evaluation of Grouping Players

The most important metric to consider validating the grouping of players are their salaries based on past IPL season data. Each grouping of players should have similar metrics atleast for some features. In this case, the metrics for the players in each cluster were observed, and by eyeballing, we can see that the players have similar metrics. If we are observing similarities between the player metrics, the clustering is effective.

Evaluation of Player Grouping- Batting

Used all steps which are mentioned in methodology section above, following are the groups of batsmen which have similar skill sets. Created two cluster groups; one by doing PCA and without PCA (by using two different main batsmen past score)

Group 1

Player	Total Runs	Batting Avg	Strike Rate
Abhishek Sharma	241	24.5	137.1
David Miller	411	29.5	118.4
Deepak Hooda	412	37.4	120.3
Dwayne Bravo	275	25.4	163.5
Eoin Morgan	551	26.4	117.0
Evin Lewis	581	25.2	131.0
Jason Roy	270	30.0	125.8
Mandeep Singh	562	25.8	124.8
Nicholas Pooran	606	23.7	146.2
Parthiv Patel	526	28.6	139.8
Pat Cummins	239	25.9	147.1
Rahul Tewatia	486	25.2	119.8
Sam Curran	337	21.9	165.9
Saurabh Tiwary	218	39.1	123.7
Shimron Hetmyer	517	25.2	146.4
Shivam Dube	399	20.2	121.1
Venkatesh Iyer	370	41.1	128.5
Vijay Shankar	611	27.3	120.6
Wriddhiman Saha	553	29.6	128.8
Yusuf Pathan	300	21.1	109.4

Group 2

Player	Total Runs	Batting Avg	Strike Rate
Chris Gayle	1339	36.1	140.5
David Warner	1435	44.2	128.7
Jonny Bairstow	1038	42.8	141.9
Jos Buttler	1441	40.7	151.1
Mayank Agarwal	1317	29.0	141.6
Rishabh Pant	1934	39.1	144.7
Sanju Samson	1642	33.7	145.5
Shane Watson	1252	31.0	134.4
Virat Kohli	1865	38.2	130.3

Group 5

Player	Total Runs	Batting Avg	Strike Rate
AB de Villiers	1689	43.6	158.9
Ambati Rayudu	1500	33.7	130.3
Faf du Plessis	1640	38.6	132.0
Kane Williamson	1474	41.1	127.3
Manish Pandey	1345	37.5	124.4
Nitish Rana	1383	28.1	134.5
Prithvi Shaw	1305	24.7	145.6
Quinton de Kock	1530	31.5	128.4
Rohit Sharma	1404	27.4	129.2
Shreyas Iyer	1568	34.5	119.5
Shubman Gill	1417	32.2	126.8
Suryakumar Yadav	1733	33.0	138.2

Group 3

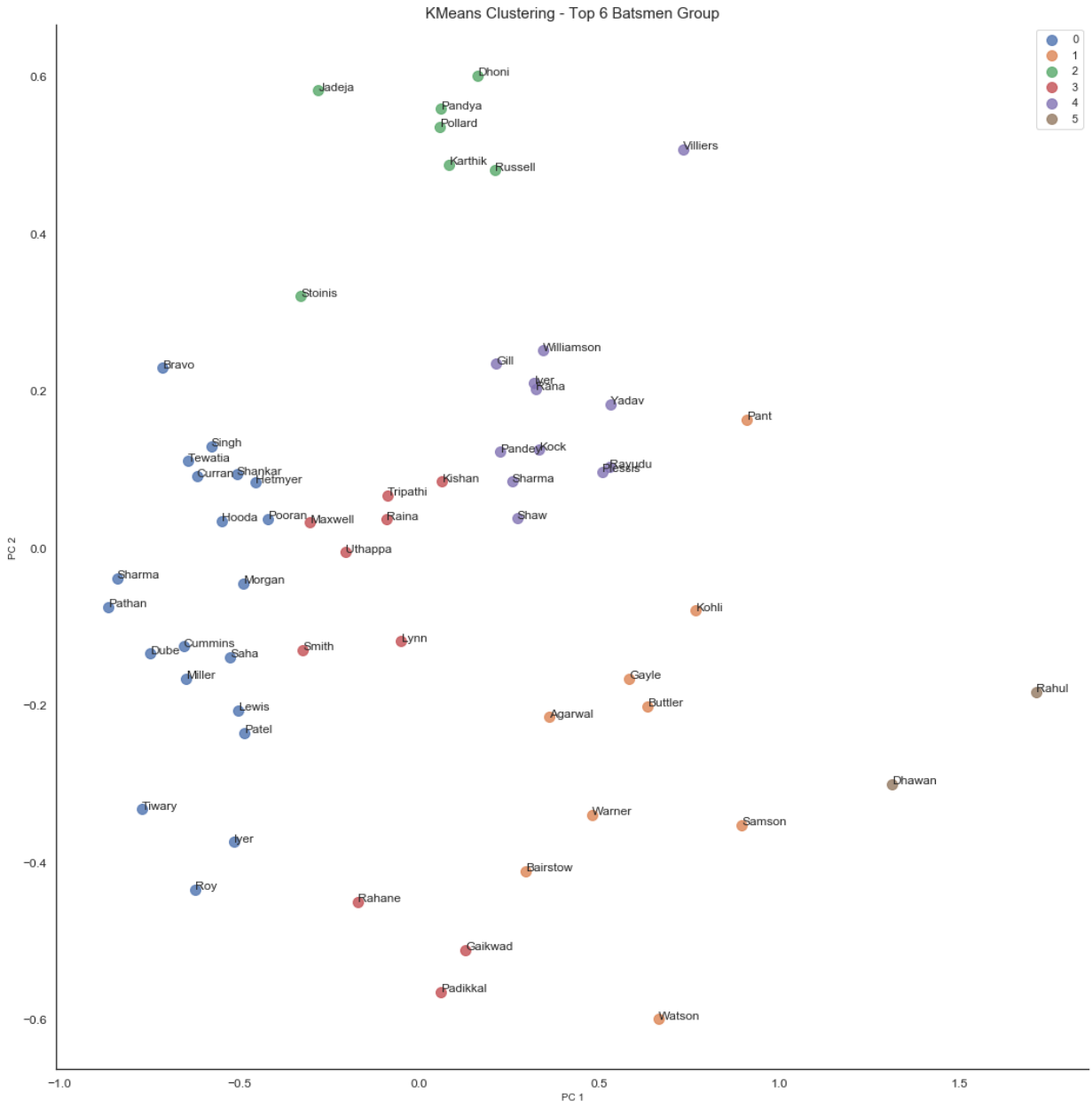
Player	Total Runs	Batting Avg	Strike Rate
Andre Russell	1126	31.1	171.6
Dinesh Karthik	1143	29.5	137.8
Hardik Pandya	1070	30.7	154.3
Kieron Pollard	925	34.5	157.4
MS Dhoni	1185	50.1	127.0
Marcus Stoinis	751	31.2	134.4
Ravindra Jadeja	654	43.8	139.5

Group 4

Player	Total Runs	Batting Avg	Strike Rate
Ajinkya Rahane	884	20.8325	115.425
Chris Lynn	945	37.626667	136.626667
Devdutt Padikkal	884	31.57	125.05
Glenn Maxwell	790	24.083333	128.936667
Ishan Kishan	1133	30.96	132.5225
Rahul Tripathi	994	24.99	130.54
Robin Uthappa	944	24.585	125.99
Ruturaj Gaikwad	839	48.175	128.485
Steve Smith	782	30.37	119.936667
Suresh Raina	988	26.26	126.47

Group 6

Player	Total Runs	Batting Avg	Strike Rate
KL Rahul	2548	56.8	140.5
Shikhar Dhawan	2223	39.1	135.5



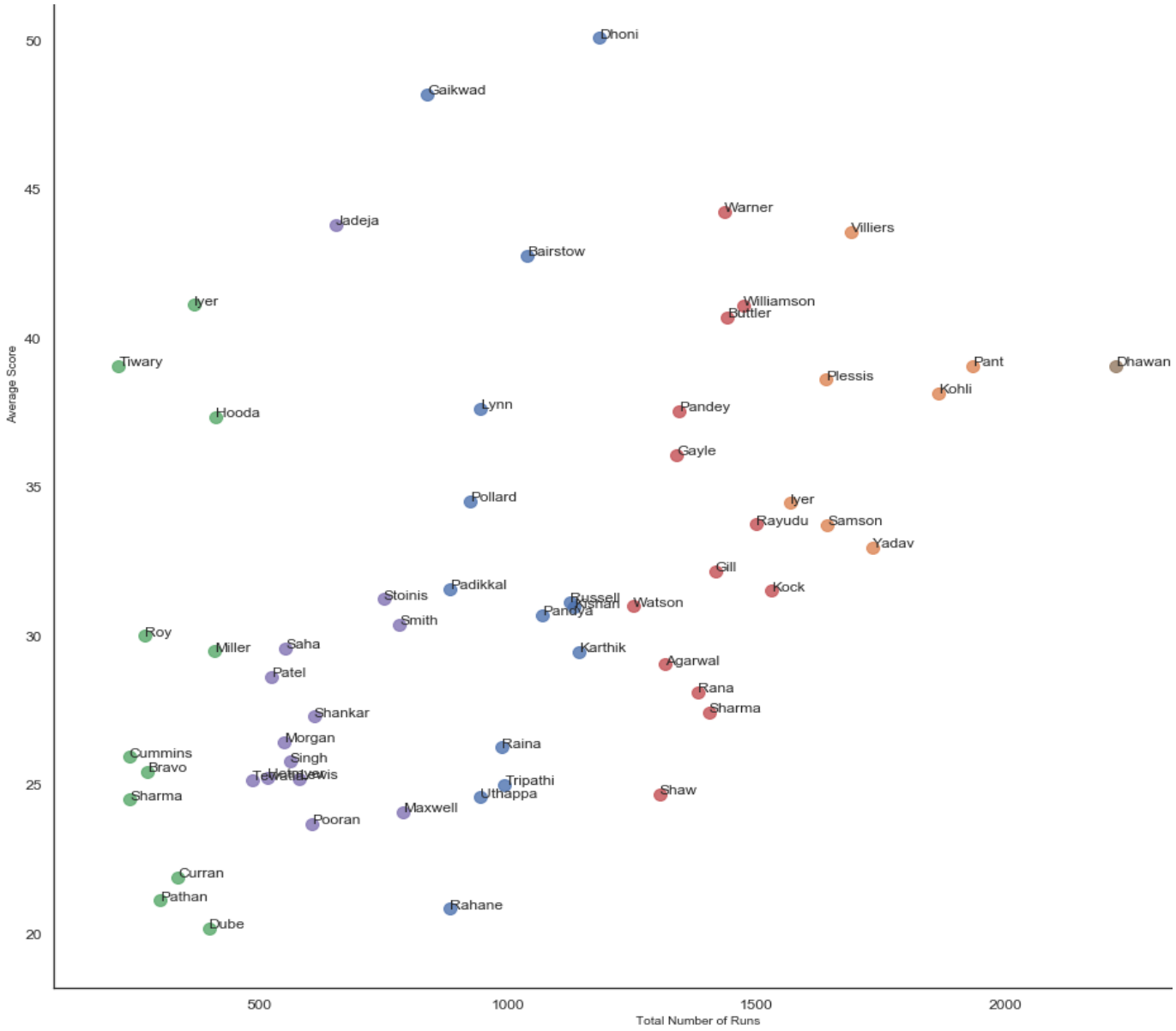
KMeans clustering of Top 6 Batsmen Group

The following observations can be made from the above batting clustering:

- Players that have a similar total number of scores have been grouped together in the same clusters, for example “KL Rahul” and “Shikhar Dhawan” are the only players who have scored more than 2000 total runs in the same group. Players who have score less total number of scores are part of one separate group.
- Players with a higher total number of scores are more right on the X axis. Those players with a high average run tend to be more up on the Y axis

Kmeans Clustering Average Score Vs. Total Number of Runs

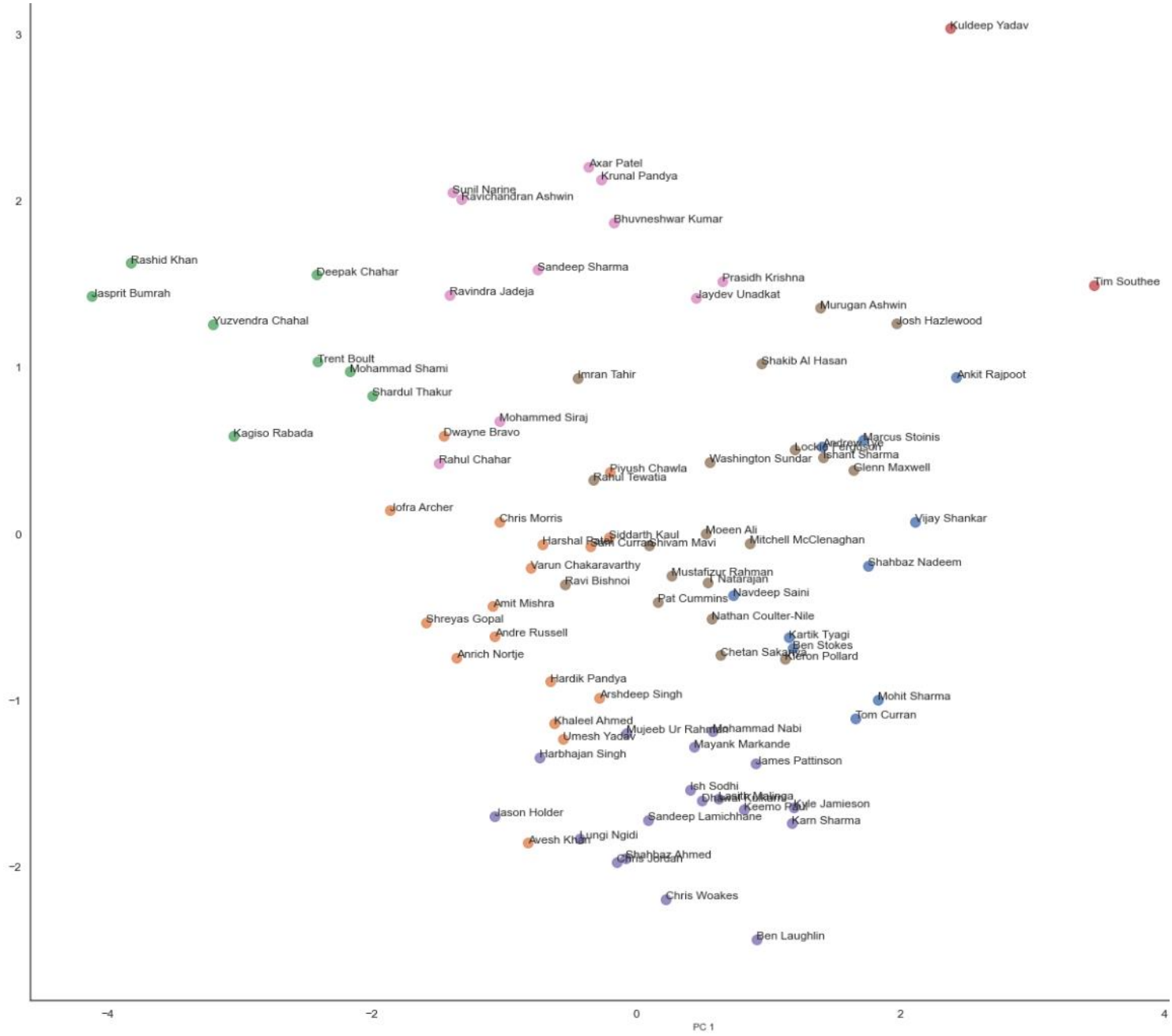
From the above clustering, we have identified that Batting "Total Run" and "Average score" tend to be prominent features of similarity between the players in each cluster. This prompted us to run Kmeans on just these attributes, and the output of the clustering is in the below scatterplot. Here we can also observe that player like "Dhawan" and "Rahul" is still in the same cluster.



Evaluation of Player Grouping - Bowling

Kmeans Clustering on All Attributes (Principal Component Analysis)

As mentioned in the methodology section for bowling, the features mentioned in the below table were used for performing K-Means Clustering. First, Principle component analysis was performed on the data and the clustering was done on all the features. The result of the clustering is mentioned in the scatterplot shown below the table:



The following observations can be made from the above clustering:

- Players that have similar bowling Strike Rate have been grouped together in the same clusters
- The clusters tend to overlap each other, which shows that some players are similar in other attributes as well such as number of wickets taken
- Players with higher number of wickets are higher up on the Y axis.
- Those players with a good bowling average tend to be more left on the X axis

This analysis would be beneficial in terms of all round performance. For example, if Jasprit Bumrah was planning on being bought by franchise A, and franchise B takes him before them, then the next fallback option would be Yuzvendra Chahal purely in terms of metrics.

The benefit of having a clustering like this gives teams quick fallback options in case their first choice players are already sold to another franchise or if the team loses the bid to another franchise.

The statistics of the players in each cluster is given below for reference:

Group 1

Player	Wickets Taken	Average	Econonmy	Strike Rate
Andrew Tye	28	48.8	10.4	27.3
Ankit Rajpoot	16	56.8	9.9	32.4
Ben Stokes	16	36.6	10.4	22.4
Kartik Tyagi	13	35.9	9.2	23.2
Marcus Stoinis	20	48.2	9.9	29.8
Mohit Sharma	9	39.3	10.4	22.5
Navdeep Saini	17	33.1	9.9	23.9
Shahbaz Nadeem	11	42.6	9.5	26.9
Tom Curran	13	38.3	10.7	21.2
Vijay Shankar	9	45.5	8.9	29.9

Group 2

Amit Mishra	32	22.2	7.2	18.6
Andre Russell	41	24.9	9.6	15.6
Anrich Nortje	34	19.4	7.3	15.9
Arshdeep Singh	30	26.5	9.3	16.8
Avesh Khan	28	17.4	9.7	10.9
Chris Morris	42	28.8	8.8	19.2
Dwayne Bravo	45	29.2	8.6	20.2
Hardik Pandya	32	24.5	9.0	16.2
Harshal Patel	44	30.3	9.1	19.9
Jofra Archer	46	22.1	7.2	18.6
Khaleel Ahmed	32	22.3	9.6	15.7
Piyush Chawla	31	34.8	9.0	23.2
Sam Curran	32	31.7	9.3	20.4
Shreyas Gopal	41	20.4	8.8	15.3
Siddarth Kaul	36	33.4	10.4	20.9
Umesh Yadav	28	22.4	9.8	14.8
Varun Chakaravarthy	36	26.9	8.4	19.7

Group 3

Player	Wkts	Avg	Econ	SR
Deepak Chahar	58	28.7	7.7	22.4
Jasprit Bumrah	84	19.5	6.9	16.9
Kagiso Rabada	70	21.1	8.1	15.6
Mohammad Shami	61	29.1	8.8	19.4
Rashid Khan	76	20.5	6.3	19.6
Shardul Thakur	55	28.7	9.0	19.1
Trent Boult	61	27.0	8.3	19.5
Yuzvendra Chahal	69	22.9	7.3	18.9

Group 7

Axar Patel	37	41.4	7.1	33.8
Bhuvneshwar Kumar	31	40.9	7.6	32.1
Jaydev Unadkat	29	45.7	9.5	29.3
Krunal Pandya	35	42.0	7.5	33.3
Mohammed Siraj	40	31.3	8.5	22.4
Prasidh Krishna	30	47.6	9.3	30.7
Rahul Chahar	41	25.7	7.4	20.9
Ravichandran Ashwin	45	36.3	7.6	28.6
Ravindra Jadeja	45	32.5	7.4	25.7
Sandeep Sharma	41	37.9	7.9	28.2
Sunil Narine	48	36.3	7.5	28.7

Group 4

Player	Wickets Taken	Average	Econonmy	Strike Rate
Krishnappa Gowtham	13	92.8	8.9	63.3
Kuldeep Yadav	22	62.7	8.2	46.5
Tim Southee	9	67.3	10.0	37.6

Group 5

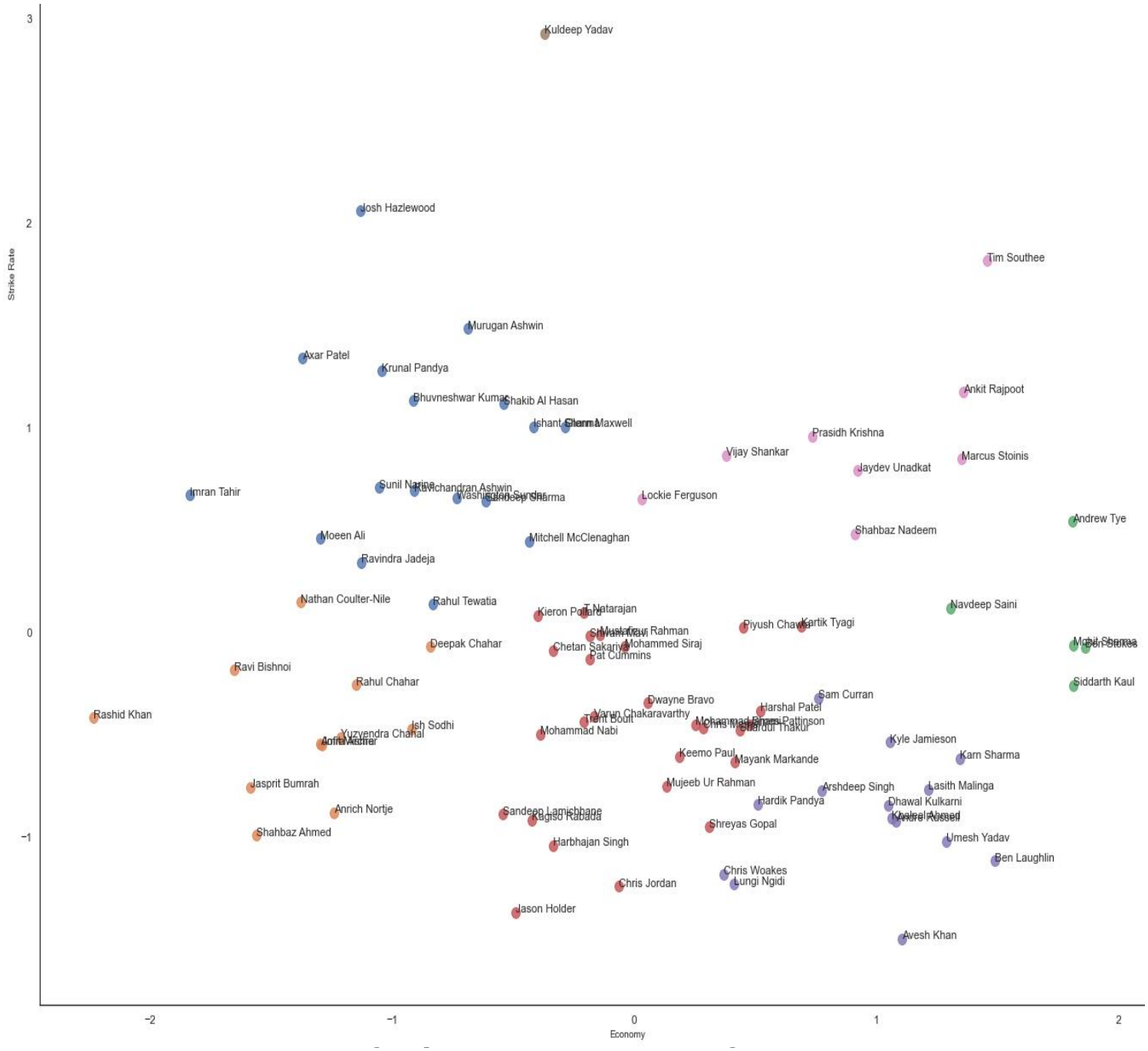
Ben Laughlin	9	23.4	10.0	14.0
Chris Jordan	13	19.3	8.5	13.0
Chris Woakes	13	20.1	8.9	13.5
Dhawal Kulkarni	10	25.7	9.6	16.2
Harbhajan Singh	23	19.4	8.2	14.6
Ish Sodhi	9	21.9	7.6	19.2
James Pattinson	11	29.1	9.0	19.4
Jason Holder	30	16.0	8.0	12.0
Karn Sharma	10	29.4	9.9	18.0
Keemo Paul	9	26.3	8.7	18.1
Kyle Jamieson	9	29.9	9.6	18.7
Lasith Malinga	16	27.4	9.8	16.8
Lungi Ngidi	25	19.2	8.9	13.1
Mayank Markande	16	27.8	9.0	17.9
Mohammad Nabi	11	28.6	8.1	19.0
Mujeeb Ur Rahman	19	24.7	8.7	16.9
Sandeep Lamichhane	13	21.3	8.0	15.8
Shahbaz Ahmed	9	17.6	7.0	15.0

Group 6

Chetan Sakariya	14	30.4	8.2	22.3
Glenn Maxwell	11	42.6	8.2	31.1
Imran Tahir	35	33.0	6.7	28.4
Ishant Sharma	14	41.3	8.1	31.1
Josh Hazlewood	12	45.3	7.4	39.5
Kieron Pollard	9	33.3	8.1	23.7
Lockie Ferguson	21	44.5	8.6	28.2
Mitchell McClenaghan	17	35.5	8.1	26.6
Moeen Ali	16	32.3	7.2	26.7
Murugan Ashwin	19	47.3	7.8	34.9
Mustafizur Rahman	21	32.0	8.4	22.9
Nathan Coulter-Nile	12	29.7	7.1	24.2
Pat Cummins	21	30.2	8.3	21.9
Rahul Tewatia	26	31.2	7.7	24.1
Ravi Bishnoi	24	24.9	6.9	21.5
Shakib Al Hasan	20	42.3	8.0	32.0
Shivam Mavi	25	32.9	8.3	22.8
T Natarajan	18	33.0	8.3	23.8
Washington Sundar	19	35.8	7.8	28.3

Kmeans Clustering on Economy Rate vs Bowling Strike Rate

From the above clustering, I have identified that Bowling Strike Rate and Economy Rates tend to be prominent features of similarity between the players in each cluster. This prompted us to run Kmeans on just these attributes, and the output of the clustering is in the below scatterplot:



- Players who have a lower strike rate (they take a smaller number of balls per wicket) are on the lower side of the y-axis
- Players who have lower economy rates are more towards the left side of the X-axis
- Those players who are more near towards the origin tend to be the most preferred players (light brown cluster) as they are both economical and wicket takers
- The farther from the origin, the worse the average

If a straight line of slope 1 is drawn through the origin, it represents the balance between strike rate and economy rate. A team should choose some extreme wicket takers, some balanced bowlers, and bowlers that can keep the opposition from scoring (economy rate). Hence this chart is very effective in deciding players to go for in the auction based on skillset.

Evaluation of Player Auction Price Prediction

Each model that would be used in predicting the auction price of players would be trained and assessed based on scores of cross validation on the 2018,2019,2020 ipl seasons as the dataset itself is small with only about 300 observations for bowling and about 300 for batting. The error in each model would be determined by the Auction Price of the Player-Model Predicted vs. Auction Price of the Player. Once the best performing model is chosen, it is evaluated on the complete and player salaries are predicted. The mean squared error is taken for each model. The error produced on the test dataset is a measure of evaluation of how well our model performed. If the error is within a reasonable threshold, then the model can be applied in future auctions given player metrics. Predictions based on performing Linear Regression, Support Vector Regression, Regression Tree, K-Nearest Neighbor, and Random Forest (only for bowling) are evaluated against each other. The model that produces the least Mean Squared Error is the model that should be chosen for future predictions.

Evaluation of Price Prediction - Batting

We used all steps which are mentioned in the methodology section for batting above to build different regression models and also predicted the auction prices. Below are list of a few players (Batsmen) which we selected to show the actual vs predicted prices for each model.

Playes	Actual Price	Linear Reg Prediction	SVR Prediction	RT Prediction	KNN Prediction
AB de Villiers	$1,718,750	$1,866,408	$1,650,879	$944,007	$1,070,834
AD Russell	$1,328,130	$1,203,212	$1,396,102	$944,007	$913,797
CH Gayle	$312,500	$1,024,844	$380,726	$944,007	$886,578
DA Warner	$1,953,130	$1,052,668	$1,885,257	$2,148,440	$759,240
KD Karthik	$1,156,250	$1,120,913	$1,087,877	$944,007	$999,632
MP Stoinis	$827,225	$957,642	$895,369	$944,007	$722,754
MS Dhoni	$2,343,750	$1,366,208	$2,275,616	$2,148,440	$1,181,289
RG Sharma	$2,343,750	$1,023,126	$563,981	$944,007	$799,731
S Dhawan	$812,500	$852,656	$880,649	$812,500	$948,958
SR Watson	$625,000	$769,811	$556,762	$944,007	$996,818
V Kohli	$2,656,250	$1,200,251	$2,274,236	$2,239,583	$1,051,042

We can see above that a few players like “MS Dhoni”, “RG Sharma” & “V Kohli”, their prediction price are not matched with few models, these are basically due to data which we took from only 2018 till 2021. Also these players are very popular in IPL and have better records in international T20 cricket as well, apart from this these players also do the captaincy role which is not considered in our regression model. We can also see that most of the models predict the salaries in a similar range. In terms of model evaluation, the Mean Squared Errors are observed for each model and are given below:

Regression Type	Mean Square Error
SVR	88837274004
Regression Tree	206240659846
Linear	340958818595
KNN	386682835719

The support vector regression and Regression Tree are giving the best results. The support vector regressor model is using a nonlinear kernel, and hence has better predictive power. Cross Validation techniques for validation compensate for possible overfitting due to the nature of the dataset which is small. Regression Tree also has a max depth of 3, to prevent possible overfitting.

Evaluation of Price Prediction -Bowling

We used all steps which are mentioned in the methodology section for batting above to build different regression models and also predicted the auction prices. Below are list of a few players(Bowlers) which we selected to show the actual vs predicted prices for each model.

	Player	ActualSalary	LinearRegressionSalaries	SVR_Regression_Salaries	RegressionTreeSalaries	KNNPredictions	RandomForestRegressor
1	A Nortje	28600	272735	90428	79642	637215	216553.4
4	AD Russell	1328130	803302	900209	1328130	615270	909032.7
7	AR Patel	714300	758312	766195	856984	740347	754992.2
8	Arshdeep Singh	28600	532808	361267	406393	718297	363393.5
9	Avesh Khan	109380	712222	747406	856984	763256	391880.1
...	...	...	...	...	...	...	...
375	SN Thakur	406250	700160	691272	955777	657685	688623.1
376	SP Narine	1953130	939506	722721	856984	833663	1377229.6
379	TA Boult	343750	573359	599038	406393	741115	543240.6
381	UT Yadav	656250	665687	678434	856984	772169	739693.8
385	YS Chahal	937500	893413	736241	856984	787516	845351.6

There is a lot of variation for some players like Avesh Khan, and Arshdeep Singh because they are relatively new players who only have experience playing 1-2 seasons and were some of the top players in the IPL. The predictions are skewed higher for those players for that reason as compared to their salary , which are low due to inexperience . Some players like AD Russel are overpaid compared to the predictions made by the models because they are traditionally star players who have performed better in editions before IPL 2018. They also have a brand value associated with themselves, which also results in a Salary higher than expected.

With regard to the models, the MSE's are listed below:

Regression Type	Mean Square Error
SVR	240524647618
Regression Tree	209276733784
Linear	301708935076
KNN	309377133784
Random Forrest	329273233784

The best performing models are random forest model, regression tree, and SVR. . The support vector regressor model is using a nonlinear kernel, and hence has better predictive power. Cross Validation techniques for validation compensate for possible overfitting due to the nature of the dataset which is small. Regression Tree also has a max depth of 4, to prevent possible overfitting. Random forest takes all possibilities into account , thereby resulting in good performance.

Future Tasks for Improvement

- Some players have a phenomenal international and domestic record when they play for their respective country and state, but have poor records/better records in IPL franchise level cricket. This skews their predicted salaries higher/lower. Taking into account their performances outside the IPL may lead to better results.
- Some players are considered as all-rounders . They can bat and bowl equally well. A separate domain should be considered for them as they are valuable assets to any team. Their salaries would also be predicted on attributes from both batting and bowling datasets
- Some players hold positions of power which may not reflect in terms of stats. For example, some team captains have poor records for a season, but since they are the captain of the franchise, they are paid more. This also needs to be taken into consideration when accounting for salary predictions.
- IPL season data from editions 2008-2017 were not considered in this analysis. Taking into records from these editions could have potentially explained why some players prices were skewed higher than normal due to amazing performances in the past versus currently

Team Work Distribution

All members of the team contributed equally to the best of their ability. Individual responsibilities are listed below:
Prakash- K-Means on Batting Data, Price Prediction on Batting Data, Documentation Preparation and Formatting
Rajeev- Data Collection, K-Means on Bowling Data, Price Prediction on Bowling Data, Documentation Preparation and Formatting.

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